Calculus Review

Q: Why to review **calculus**? In numerical mathematics??

A: When developing numerical schemes we will use theorems from calculus to guarantee that our algorithms make sense.

Key concepts from calculus:

- Limits
- Continuity
- Convergence
- Differentiability
- Extreme Value Theorem
- Intermediate Value Theorem
- Mean Value Theorem
- Taylor's Theorem

Limit/Convergence

Definition (Limit)

A function f defined on a set $X \subset \mathbb{R}$ has the **limit** L at x_0, written

$$\lim_{x \to x_0} f(x) = L$$

if for any sequence (x_n) approaching x_0 a sequence $(f(x_n))$ approaches L.

Definition (Continuity (at a point))

Let f be a function defined on a set $X \subset \mathbb{R}$, and $x_0 \in X$. Then f is **continuous at** x_0 if

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Continuity/Convergence

Definition (Continuity (in an interval))

A function f is **continuous on a set** $X \subset \mathbb{R}$ (i.e., $f \in C(X)$) if it is continuous at each point $x \in X$.

Differentiability

Theorem

If f is a function defined on a set $X \subset \mathbb{R}$ and $x_0 \in X$, then the following statements are ***equivalent****:*

(a) *f is continuous at x_0*

(b) *If $\{x_n\}_{n=1}^{\infty}$ is any sequence in X converging to x_0, then $\lim_{n\to\infty} f(x_n) = f(x_0)$.*

Definition

Let f be a function defined on an open interval containing x_0 (i.e., $x_0 \in (a, b)$). Then f is **differentiable at** x_0 if

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \text{ exists.}$$

If the limit exists, we call $f'(x_0)$ a derivative of f at x_0.

Continuity

Theorem (Differentiability $\Rightarrow$ Continuity)

If f is differentiable at x_0, then f is continuous at x_0.

Extreme Value Theorem

Theorem (Extreme Value Theorem)

If $f \in C[a, b]$ *then*

$$\exists m, M \in [a, b] \forall x \in [a, b] : f(m) \leq f(x) \leq f(M).$$

I.e., f *attains its* ***minimum*** *at* m *and* ***maximum*** *at* M*.*
Moreover, if f *is differentiable on* (a, b) *then the numbers* m, M *occur either at endpoints* a, b *or where* $f'(x) = 0$*.*

Mean and Intermediate Value Theorem

Theorem (Mean Value Theorem)

If $f \in C[a,b]$ and f is differentiable on (a,b), then $\exists c \in (a,b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Theorem (Intermediate Value Theorem)

If $f \in C[a,b]$ and $K \in (f(a), f(b))$, then there ex ts a number $c \in (a,b)$ for which $f(c) = K$.

Taylor's Theorem

Theorem (Taylor's Theorem)

Suppose $f \in C[a,b]$, $f^{(n+1)}$ *exists on* (a,b) *and* $x_0 \in [a,b]$. *Then* $\forall x \in (a,b)$, $\exists \xi \in (x_0, x)$ *with*

$$f(x) = P_n(x) + R_n(x)$$

where
P_n *is called the* ***Taylor polynomial*** *of degree* n, *and* $R_n(x)$ *is the* ***remainder term*** *(truncation error).*

This theorem is extremely important for numerical analysis:
Taylor expansion is a fundamental step in the derivation of many of the algorithms we see in this class.

Computer Arithmetic and Finite Precision

Finite Precision: A 64-bit real number, double

The *Binary Floating Point Arithmetic Standard 754-1985* (IEEE - The Institute for Electrical and Electronics Engineers) standard specified the following layout for a 64-bit real number:

$$\mathbf{s c_{10}\, c_9 \ldots c_1\, c_0\, m_{51}\, m_{50} \ldots m_1\, m_0}$$

where

Symbol	Bits	Description
s	1	The sign bit: 0 = positive, 1 = negative
c	11	The characteristic (exponent)
m	52	The mantisa

$$r = (-1)^s 2^{c-1023}(1+m), \quad c = \sum_{k=0}^{10} c_k 2^k, \quad m = \sum_{k=0}^{51} \frac{m_k}{2^{52-k}}$$

Finite Precision: Examples

$$r = (-1)^s 2^{c-1023}(1+m), \quad c = \sum_{k=0}^{10} c_k 2^k, \quad m = \sum_{k=0}^{51} \frac{m_k}{2^{52-k}}$$

Remarks:

- $2^{10} = 1024$ and $(11111111111)_2 = 2047$.
- We cannot represent an exact zero!

Example 1: 3.0

0 10000000000 1000000000000000000000000000000000000000000000000

$$(-1)^0 \cdot 2^{2^{10}-1023} \cdot \left(1 + \frac{1}{2}\right) = 1 \cdot 2^1 \cdot \frac{3}{2} = 3.0$$

Finite Precision: Examples

$$r = (-1)^s 2^{c-1023}(1+m), \quad c = \sum_{k=0}^{10} c_k 2^k, \quad m = \sum_{k=0}^{51} \frac{m_k}{2^{52-k}}$$

Example 2: The Smallest Positive Real Number

0 00000000000 0000000000000000000000000000000000000000000000000001

$$r = (-1)^0 \cdot 2^{0-1023} \cdot \left(1 + \frac{1}{2^{52}}\right) = (1 + 2^{-52}) \cdot 2^{-1023} \cdot 1 \approx 10^{-308}$$

Example 3: The Largest Positive Real Number

0 11111111110 1111111111111111111111111111111111111111111111111111

$$r = (-1)^0 \cdot 2^{1023} \cdot \left(1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{51}} + \frac{1}{2^{52}}\right) \quad = 2^{1023} \cdot \left(2 - \frac{1}{2^{52}}\right) \approx 10^{308}$$

Finite Precision: Consequences

- There are gaps in the floating-point representation. I.e., any number in the interval

$$\left[3.0, 3.0 + \frac{1}{2^{52}}\right)$$

  is represented by value 3.0.
- Floating point "numbers" represents intervals!

Quantifying the Error

Let p^* be an approximation to p, then

Definition (The Absolute Error)

$$|p - p^*|$$

Definition (The Relative Error)

$$\frac{|p-p^*|}{|p|}, \quad p \neq 0$$

Sources of Numerical Errors - Roundoff Errors (Rounding and Truncating) I

Examples in 5-digit arithmetic

Rounding 5-digit arithmetic:

$$(96384 + 26.678) - 96410 =$$
$$(96384 + 00027) - 96410 =$$
$$96411 - 96410 = 1.0000$$

Truncating 5-digit arithmetic:

$$(96384 + 26.678) - 96140 =$$
$$(96384 + 00026) - 96410 =$$
$$96410 - 96410 = 0.0000$$

Sources of Numerical Errors - Roundoff Errors (Rounding and Truncating) II

Rearrangement changes the result:

$$(96384 - 96410) + 26.678 = -26.000 + 26.678 = 0.67800$$

Numerically, order of computation matters!

Algorithms

Definition (Algorithm)

An **algorithm** is a procedure that describes, in an **unambiguous** manner, a finite sequence of steps to be performed in a specific order.

In this class, the objective of an algorithm is to solve a problem or approximate a solution to a problem.
Algorithms work very similarly to the meal recipes.

Key Concepts for Numerical Algorithms – Stability

Definition (Stability)

An algorithms is said to be **stable** if small changes in the input, generate small changes in the output.

At some point we need to quantify what "small" means!
If an algorithm is stable for a certain **range** of initial data, then it is said to be **conditionally stable**.

Key Concepts for Numerical Algorithms – Error Growth

Suppose $E_0 > 0$ denotes the initial error, and E_n represents the error after n operations.

If $E_n \approx C \cdot E_0 \cdot n$ (for a constant C which is independat of n), then the growth is **linear**.

If $E_n \approx C^n \cdot E_0$, $C > 1$, the the growth is **exponential** – in this case the error will dominate very fast (undesirable scenario).

Linear error growth is usually unavoidable, and in the case where C and E_0 are small the results are generally acceptable. – **Stable algorithm.**

Exponential error growth is unacceptable. Regardless of the size of E_0 the error grows rapidly. – **Unstable algorithm.**

Reducing the Effects of Roundoff Error

The effects of roundoff errors can be reduced by using higher-order-digit arithmetic such as the double or multiple-precision arithmetic available on most computers.

Disadvantages in using double precision arithmetic are that it takes more computation time and **the growth of the roundoff error is not eliminated but only postponed**.

Sometimes, but not always, it is possible to reduce the growth of the roundoff error by restructuring the calculations.

Key Concepts - Rate of Convergence

Definition (Rate of Convergence)

Suppose the sequence $\beta = \{\beta_n\}_{n=1}^{\infty}$ converges to zero, and $\alpha = \{\alpha_n\}_{n=1}^{\infty}$ converges to a number α.
If $\exists K > 0 : |\alpha_n - \alpha| < K \cdot \beta_n$, for n large enough, then we say that $\{\alpha_n\}_{n=1}^{\infty}$ converges to α with
a **Rate of Convergence** $\mathcal{O}(\beta_n)$ ("Big Oh of β_n").
We write

$$\alpha_n = \alpha + \mathcal{O}(\beta_n)$$

Note: The sequence $\beta = \{\beta_n\}_{n=1}^{\infty}$ is usually chosen to be

$$\beta_n = \frac{1}{n^p}$$

for some positive value of p.

Rate of Convergence: Example

Consider the sequence (as $n \to \infty$)

$$\alpha_n = \sin\left(\frac{1}{n}\right) - \frac{1}{n}.$$

Then $\alpha_n = \mathcal{O}\left(\frac{1}{n^3}\right)$.

Generalizing to Limits of Functions

Definition (Rate of Convergence)

Suppose

$$\lim_{h\to 0} G(h) = 0, \text{ and } \lim_{h\to 0} F(h) = L.$$

If $\exists K > 0 \forall h < H$ (for some $H > 0$):

$$|F(h) - L| \leq K|G(h)|$$

then

$$F(h) = L + \mathcal{O}(G(h)).$$

We say that $F(h)$ converges to L with a **Rate of convergence** $\mathcal{O}(G(h))$.

Note: Usually we consider $G(h) = h^p$ for some positive p.

Rate of Convergence: Example

Consider the function

$$\alpha(h) = \sin(h) - \frac{1}{h}.$$

Then $\alpha(h) = \mathcal{O}(h^3)$.